Quantitative Financial Risk Management

Assignment 1: Portfolio VaR & ES

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Contents

Introduction **2**

Data and Portfolio Construction **3**
Asset selection . . . 3
Data cleaning and missing observations . . . 3
Loan construction . . . 4
Portfolio weights . . . 4

Individual Asset VaR and ES **7**
Method 1 — Variance-Covariance (Normal) . . . 7
Method 2 — Variance-Covariance (Student-t) . . . 7
Method 3 — Historical Simulation . . . 8
Method 4 — GARCH(1,1) . . . 9
Method 5 — Filtered Historical Simulation (EWMA) . . . 10
Overall comparison . . . 10
Effect of sample choice . . . 11

Portfolio VaR and ES **12**
Method 1 — Variance-Covariance (Normal) . . . 12
Method 2 — Variance-Covariance (Student-t) . . . 13
Method 3 — Historical Simulation . . . 13
Method 4 — GARCH(1,1)-CCC . . . 13
Method 5 — Filtered Historical Simulation (EWMA) . . . 14
Effect of sample choice . . . 14
Conclusion . . . 15

Backtesting of VaR and ES **16**
Backtesting setup . . . 16
Backtesting of VaR: frequency of violations . . . 16
Backtesting of ES: predicted ES versus actual average shortfall . . . 19
Dependence and clustering of VaR violations . . . 21
Overall conclusion and recommendation . . . 22

Multi-Day Value-at-Risk Analysis **23**
Non-Overlapping h-Day Return Construction . . . 23
Historical Simulation VaR . . . 23
Square-Root-of-Time Rule . . . 24
Comparison Framework . . . 24
Results . . . 25
Violation Analysis and Binomial Test . . . 26
Independence of Violations . . . 27

Stress Testing **28**
Deterministic Stress Testing Results . . . 28
Distribution-Based Stress Testing Results . . . 29
Interest Rate Sensitivity Analysis . . . 29

Introduction

This report presents a quantitative analysis of portfolio Value-at-Risk (VaR) and Expected Shortfall (ES) for a multi-asset portfolio over the period January 2006 to December 2013. The portfolio is constructed to ensure simultaneous exposure to equity risk, interest rate risk, and exchange rate uncertainty, combining assets across different sectors and currency areas. The estimation period is chosen to encompass the Financial Crisis of 2008–2009. This choice allows us to examine how different risk models perform both in calm market conditions and under acute stress, and to assess the sensitivity of VaR and ES estimates to the inclusion or exclusion of crisis observations.

After describing the data construction and portfolio weights, we compute individual asset VaR and ES using five methods: the Variance-Covariance approach under normality and Student-t assumptions, Historical Simulation, GARCH(1,1), and Filtered Historical Simulation with EWMA. These methods are then applied at the portfolio level. Subsequently, we backtest the portfolio VaR and ES, extend the analysis to multi-day horizons, and conclude with a stress testing exercise evaluating the portfolio's resilience under extreme shocks.

Data and Portfolio Construction

Asset selection

In this report, $N = 7$ assets are selected over the period from January 1, 2006 to December 31, 2013. The data are downloaded via Yahoo Finance, with the exception of the loan rate, which is obtained from the Federal Reserve Economic Data (*FRED*), where the 3-month Treasury Bill rate (*DTB3*) is used. The portfolio, which consists of three stocks, two market indices, an exchange rate and a loan component, is constructed to ensure exposure to stock price variation, interest rate risk, and exchange rate uncertainty.

The three stocks included are ExxonMobil (XOM) and Shell (SHEL), representing the global energy sector, and ING Group (INGA.AS), representing the European financial sector. ING Group is particularly relevant given its severe exposure to the 2008-2009 financial crisis. This combination of assets ensures exposure to different sectors, geographies, and risk factors simultaneously, making the portfolio suitable for a comprehensive VaR and ES analysis. The two market indices obtained in this report are the S&P 500 (ˆGSPC) and AEX (ˆAEX). Since the portfolio contains both USD-denominated assets (ExxonMobil, Shell and S&P 500) and EUR-denominated assets (ING and AEX), returns are retained in their local currency. The EUR/USD (EURUSD=X) exchange rate is included to account for the currency risk arising from holding assets across two different currency areas. An overview of the assets is shown in Table 1.

Table 1: Overview of portfolio assets

Asset	**Ticker**	**Type**	**Currency**	**Exchange**
ExxonMobil	XOM	Stock	USD	NYSE
Shell	SHEL	Stock	USD	NYSE
ING Group	INGA.AS	Stock	EUR	Euronext Amsterdam
S&P 500	ˆGSPC	Index	USD	–
AEX	ˆAEX	Index	EUR	Euronext Amsterdam
EUR/USD	EURUSD=X	FX rate	EUR/USD	–
Loan (DTB3)	–	Loan	USD	FRED

Data cleaning and missing observations

Before applying any cleaning procedures, the raw series were inspected for missing values. The results are summarized in Table 2

Table 2: Missing observations per series before and after cleaning

Series	**Missing (raw)**	**% missing**	**Max streak**	**Remaining after clean**
XOM	74	3.55%	2	0
SHEL	74	3.55%	2	0
ˆGSPC	74	3.55%	2	0
ˆAEX	41	1.97%	2	0
INGA.AS	36	1.73%	2	0
EURUSD=X	24	1.15%	11	0
DTB3	84	4.02%	1	0
Total rows dropped				23 (2,086 → 2,063)

We find that the stocks and market indices share a maximum consecutive missing streak of two days, most likely caused by public holidays falling on different days across exchanges. The EUR/USD exchange rate has a maximum streak of eleven days, reflecting the different trading calendar of the foreign exchange markets.

Missing observations are handled using forward filling, which carries the last observed price forward in the case of a missing observation. This approach is often used in finance for synchronizing financial time series with different trading calendars, as it allows alignment without introducing artificial jumps in prices. Given the difference in maximum missing observation streaks, we decided to handle this differently for stocks, market indices, the loan, and the exchange rate.

For stocks and market indices, we limited forward filling to one day. This reflects a conservative approach; short gaps can be attributed to asynchronous trading days across markets, while longer gaps may lead to artificial price consistency and thus underestimate return volatility. Therefore, observations with longer missing sequences were removed. The DTB3 loan only consisted of one-day gaps, so it was similarly forward-filled with the same maximum of one day.
For the EUR/USD exchange rate, a less restrictive forward-filling rule is applied. Unlike equity markets, foreign exchange markets trade continuously, and missing observations in the dataset are primarily due to data availability issues. As a result, we applied more flexible five-day forward filling to preserve information and avoid unnecessary data loss. After applying these procedures, any remaining missing values were removed, resulting in a clean and synchronized dataset of 2,063 daily observations.

Loan construction

Since the DTB3 rate varies over time, the loan is treated as a floating rate instrument. A constant credit spread of 1% is added to the short-term risk-free rate to represent the borrowing costs above the risk-free benchmark. The 1% spread is consistent with typical borrowing costs of institutions and keeps the model tractable. The resulting annualized rate is converted to a daily log return by dividing by 252 trading days as follows:

$$r_{\text{loan},t} = \ln\left(1 + \frac{r_{\text{DTB3},t} + \text{spread}}{252}\right) \tag{1}$$

Portfolio weights

To determine the portfolio weights, daily log returns r_t are computed for each asset. For all assets except the loan, whose daily log return is defined in equation 1, the close-to-close log returns are obtained as $r_t = \log\left(\frac{P_t}{P_{t-1}}\right)$, where P_t denotes the adjusted closing price on day t. In this case, simple returns are not used, as they are not additive over time. This because they can complicate the analysis of multi-period returns. In contrast, log returns are time-additive. Log returns are time-additive, meaning that the multi-period return is simply the sum of the individual one-period log returns, which facilitates aggregation over time (Tsay, 2010). Furthermore, taking log returns induces stationarity in the mean. The computed daily log returns for all time series are plotted in Figure 1.

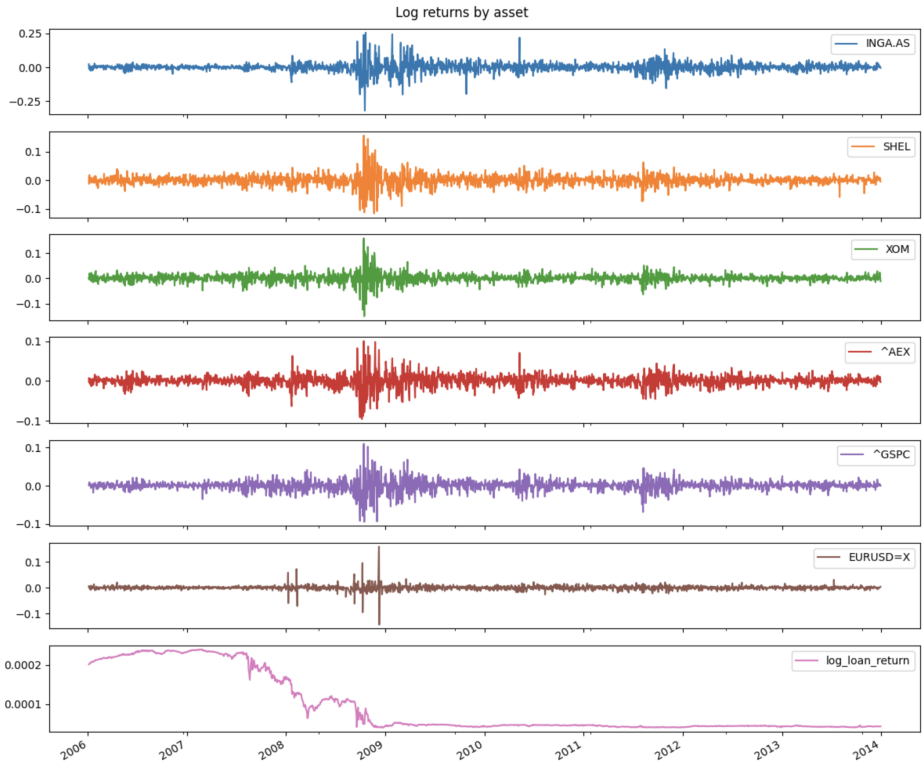

Figure 1: Daily log returns for each portfolio component individually

Tsay, R. S. (2010). Analysis of Financial Time Series. John Wiley & Sons.

The portfolio weights satisfy $\mathbf{w} = (w_1, ..., w_N)$ where $\sum w_i = 1$, and are fixed throughout the sample period. As shown in Figure 1, pronounced volatility clustering is visible across the stock and market indices. ING Group exhibits the highest volatility, with a negative return of 0.321 during the 2008-2009 financial crisis, consistent with the severe distress the banking institution faced during this period. The loan return, by contrast, shows a smooth decline from 2006 to nearly zero after 2009 and remaining constant thereafter. This directly reflects the Federal Reserve's interest rate cuts in response to the financial crisis. The EUR/USD exchange rate displays comparatively low volatility throughout the sample, as it serves primarily as a currency hedge. Based on these observations, the weight assigned to ING Group is reduced relative to the other equities, the weight assigned to EUR/USD is similarly reduced, and the weight assigned to the loan is increased. The resulting weights are summarized in Table 3.

Table 3: Portfolio weights per asset

Asset	**Ticker**	**Weight**
ExxonMobil	XOM	15%
Shell	SHEL	15%
ING Group	INGA.AS	10%
S&P 500	ˆGSPC	15%
AEX	ˆAEX	15%
EUR/USD	EURUSD=X	10%
Loan (DTB3)	–	20%
Total		**100%**

Figure 2 shows the daily log returns of the resulting fixed-weights portfolio, combining the volatility contributions of all seven assets according to the chosen weights. Pronounced volatility clustering is visible across the full sample,

with the most severe spike occurring during the 2008–2009 financial crisis, consistent with the patterns observed in the individual series.

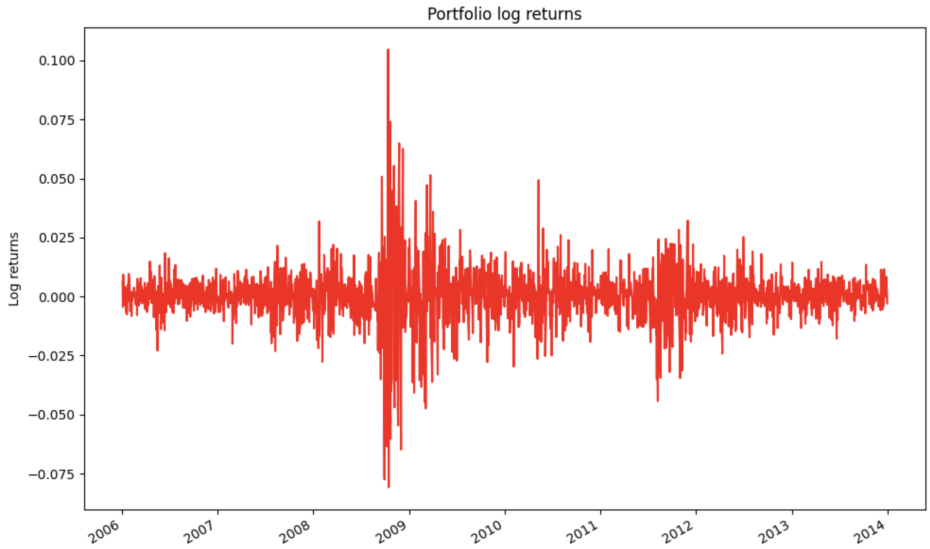

Figure 2: Daily log returns of the fixed-weights portfolio

Individual Asset VaR and ES

The VaR and ES will now be calculated using several different methods applied to our dataset. Each method will be evaluated, and the differences between them will be analysed and compared.

In this analysis, a confidence level of 99% ($\alpha = 1\%$) is used for the calculation of Value-at-Risk and Expected Shortfall for all methods. This is chosen because it is consistent with the standard practice in financial risk management, where the focus lies on extreme losses in the tail of the return distribution. Using a 99% confidence level ensures that the risk measures capture severe but plausible adverse market movements.

Method 1 — Variance-Covariance (Normal)

First we have performed the normal VCV method; this gives the lowest VaR estimates across almost all assets. As can be seen in Table 4, the equity risk factors range from 2.18% (EURUSD=X) to 8.29% (INGA.AS), with the two market indices ˆAEX (3.40%) and ˆGSPC (3.20%) sitting below the individual stocks as expected, since indices are inherently diversified and therefore exhibit lower daily return volatility than single names. The loan component (`log_loan_return`) produces a VaR of almost zero (0.0083%), reflecting that the loan return series has near-constant daily values and almost no volatility. INGA.AS stands out with a VaR of 8.29%, roughly double that of SHEL and XOM, which is consistent with ING being a financial sector stock that was particularly affected during the Global Financial Crisis. As with any application of the normal method, these figures should be treated as a lower bound because the normal distribution has thin tails and therefore tends to underestimate the probability of large losses.

Table 4: 1-day 99% VaR and ES, full sample 2006–2013. All values are expressed as percentages of portfolio value.

Asset	**Norm VaR**	**Norm ES**	***t* VaR**	***t* ES**	**HS VaR**	**HS ES**	**GARCH VaR**	**GARCH ES**	**FHS VaR**	**FHS ES**
INGA.AS	8.2906	9.4936	9.3386	14.3854	11.3571	15.1809	3.6903	4.2350	4.1952	5.4887
SHEL	4.2212	4.8396	4.7599	7.3541	5.4736	7.7846	2.3971	2.7530	2.5283	3.3528
XOM	3.7933	4.3509	4.2791	6.6186	4.6385	6.7287	2.8374	3.2592	3.1377	4.0586
ˆAEX	3.4004	3.8950	3.8313	5.9063	4.4184	6.1156	2.0456	2.3512	2.3274	2.7479
ˆGSPC	3.1969	3.6652	3.6049	5.5696	4.4274	6.0837	1.4716	1.6953	1.7301	2.2801
EURUSD=X	2.1757	2.4935	2.4526	3.7861	1.8977	3.8083	0.8707	1.0015	0.8345	1.0735
log_loan_ret	0.0083	0.0109	0.0102	0.0157	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Method 2 — Variance-Covariance (Student-t)

Based on the QQ-plots visible in Figure 3, df = 3 was selected for all equity and FX assets, while the loan return received df = 6, which is the highest available and closest to normal (consistent with its near-zero volatility and essentially flat return series that has very little tail behaviour to model). The uniform selection of df = 3 for all risk assets (the heaviest tail option available) is a strong result. It indicates that all six market series have clearly fat tails over the 2006–2013 period, which is precisely what one would expect from a sample dominated by the crisis of 2008.

The effect of this method on VaR is moderate, but the effect on ES is substantial. INGA.AS moves from a normal VaR of 8.29% to a t-VaR of 9.34%, while its ES jumps from 9.49% to 14.39%, as also visible in Table 4. The same pattern can be seen across all assets. The Student-t distribution adds more mass in the far tail beyond the VaR threshold, which the normal model misses. For risk management purposes this difference in ES is particularly important, since ES measures the expected loss once the VaR threshold has been breached.

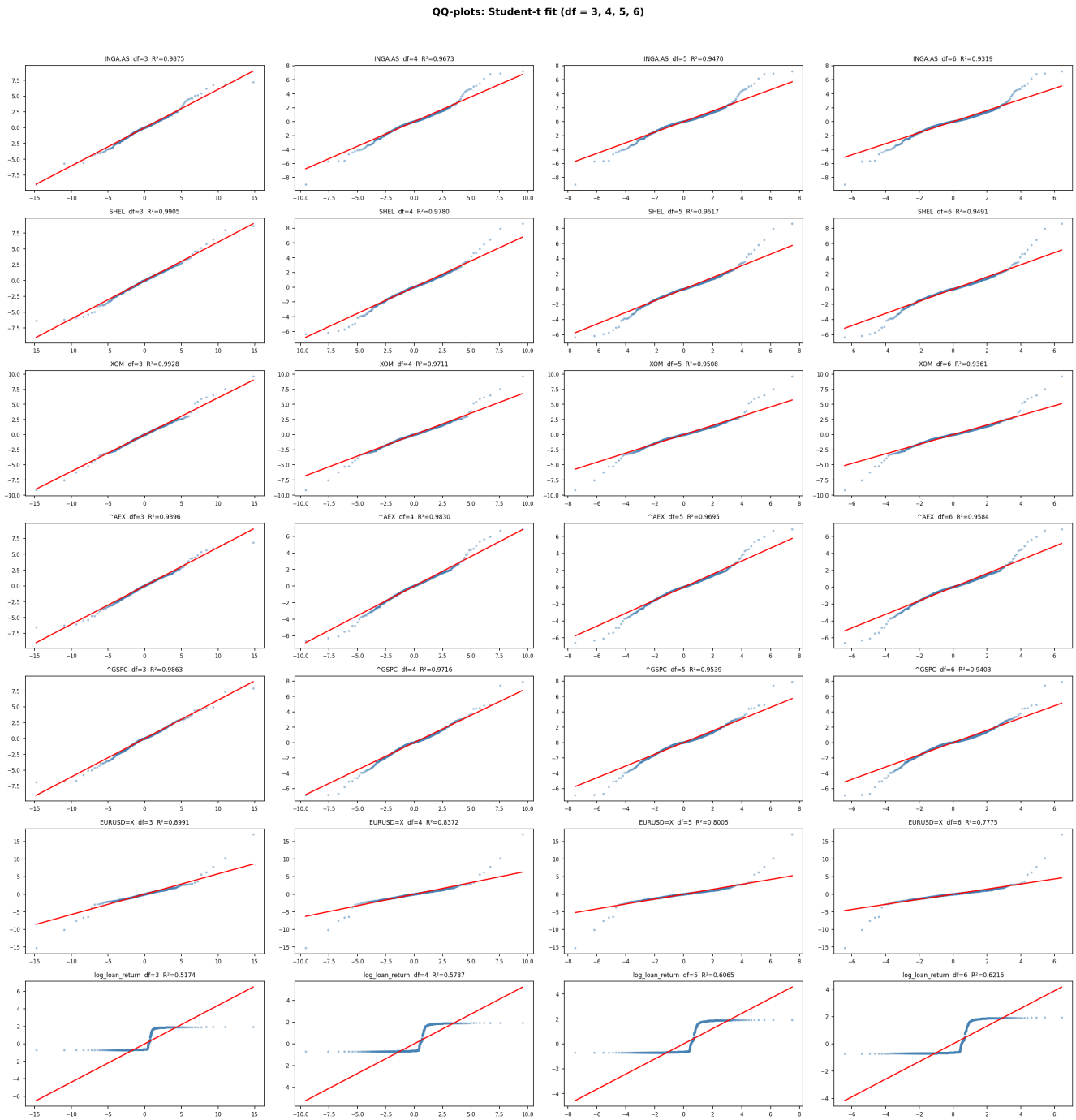

Figure 3: Normal QQ-plots for each individual portfolio component. Departures from the diagonal confirm heavy-tailed behaviour and support the selection of df = 3 for all equity and FX assets.

Method 3 — Historical Simulation

Historical Simulation makes no distributional assumption; VaR is simply the empirical 1st percentile and ES is the average of losses beyond it. Looking at Table 4, the HS estimates are the highest of any non-forward-looking methods for the equity assets. INGA.AS reaches a VaR of 11.36% and an ES of 15.18%, SHEL 5.47% and 7.78%,

and the indices around 4.4% each. These figures are largely above the normal estimates, driven directly by the extreme loss days observed during the GFC peak in late 2008 and early 2009. The loan component produces VaR and ES of exactly 0.00%, consistent with its near-constant return series.

EURUSD=X is the main exception: its HS VaR of 1.90% is slightly below its normal VaR of 2.18%. This happens because the empirical distribution of EUR/USD returns is relatively well-behaved in the body but has a few very large positive outliers, meaning the 1st percentile of losses is smaller than the normal model would predict based on the standard deviation. However, its HS ES of 3.81% remains much higher than the normal ES of 2.49%, confirming that once the tail is reached, the losses can still be large.

Method 4 — GARCH(1,1)

The GARCH results introduce time-varying volatility and produce the lowest VaR estimates of all five methods for every asset, because the 1-step ahead forecast is made from end-2013, a period of low and declining volatility after the GFC had resolved. All equity assets show high volatility persistence, with $\hat{\alpha} + \hat{\beta}$ close to one: SHEL (0.987), XOM (0.983), $\hat{AEX}$ (0.988), and $\hat{GSPC}$ (0.986). For INGA.AS and EURUSD ≈ 1.000, the reported value of 1.000 is because of rounding, as the estimated sum is actually slightly below one at higher precision. This indicates near-integrated GARCH behaviour, this implies that volatility shocks are extremely persistent and decay only very slowly, especially for INGA.AS and EURUSD.

The conditional volatility plots in Figure 4 make this very clear: all six market assets show strong spikes during the GFC window, with INGA.AS reaching annualised volatility above 250%, while SHEL and ˆGSPC rise above roughly 80–100%. These spikes then gradually decline through 2010–2013, so by end-2013 the forecast volatility is close to historical lows. This explains why GARCH produces so much lower VaR values than Historical Simulation; for example, INGA.AS has a GARCH VaR of only 3.69% versus 11.36% under HS. The loan return again shows near-zero VaR and ES, with slightly unusual GARCH parameters ($\hat{\alpha} = 0.835$, $\hat{\beta} = 0.054$) caused by its almost constant series as the optimiser finds a degenerate solution with very low persistence.

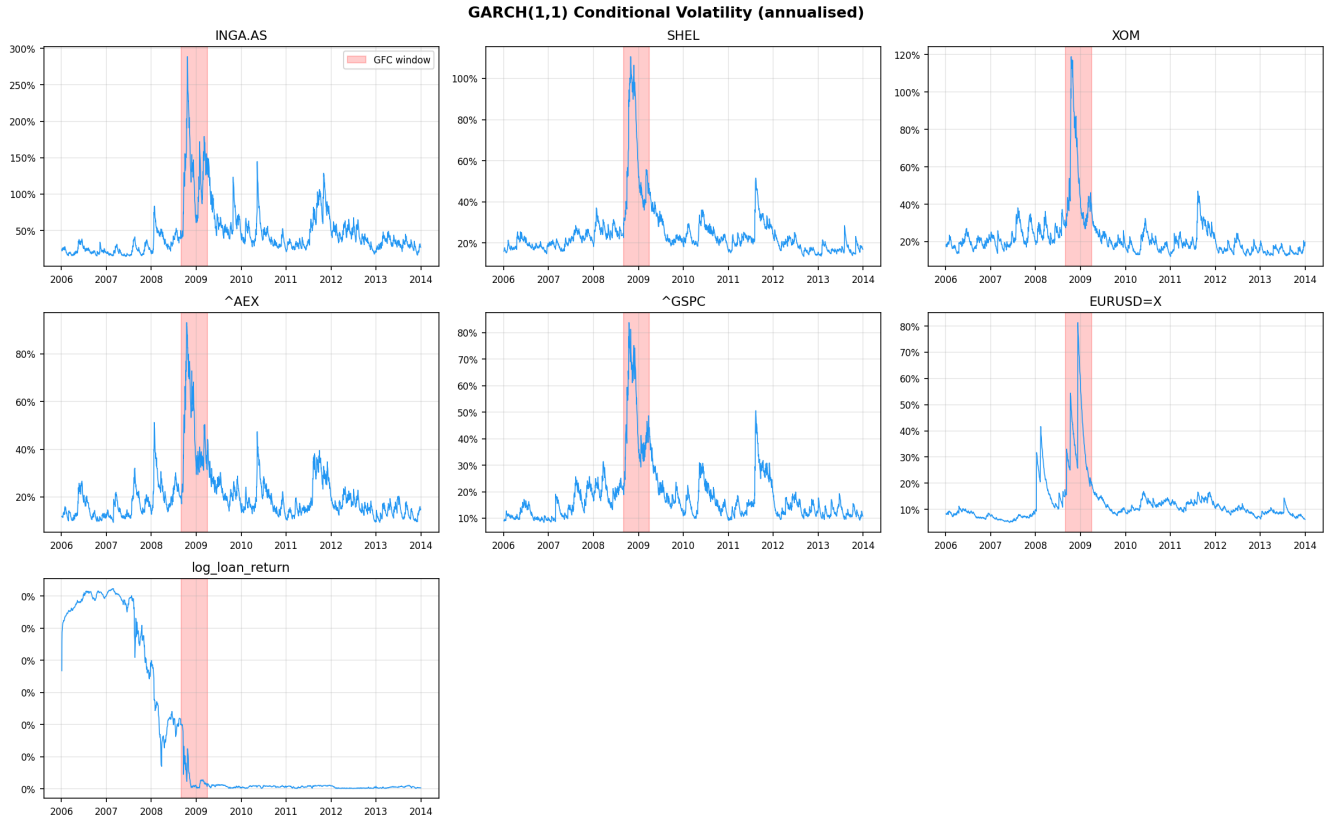

Figure 4: GARCH(1,1) conditional volatility estimates for each individual portfolio component. The pronounced spikes during the 2008–2009 GFC period and the subsequent decline towards end-2013 explain why the GARCH-based VaR estimates are substantially lower than those obtained from Historical Simulation.

Method 5 — Filtered Historical Simulation (EWMA)

The FHS-EWMA method produces estimates between GARCH and plain Historical Simulation, which is the expected outcome. INGA.AS has a VaR of 4.20% and an ES of 5.49%, well above the GARCH forecast but far below HS. The EWMA filter with $\lambda = 0.94$ downweights older crisis observations relative to plain HS, but the standardised residuals from the GFC period still contribute to the empirical tail distribution. Because end-2013 volatility is low, these residuals are rescaled downward relative to their original crisis magnitudes, which suppresses the VaR estimate compared with HS. The loan component again produces VaR and ES of 0.00%, while EURUSD=X shows a similar pattern to Historical Simulation, with FHS VaR of 0.83% falling below the normal estimate of 2.18% due to very low end-of-sample EWMA volatility.

Overall comparison

The spread between methods is much larger here than it would be in a typical calm-period sample, as the GFC dominates the 2006–2013 window. The gap between GARCH, which is state-dependent and based on the end-2013 regime, and HS, which averages over the full crisis-inclusive history, is especially large for INGA.AS. The normal model provides a useful benchmark; the Student-t method mainly raises ES by capturing fat tails; Historical Simulation is the most conservative unconditional method; and GARCH and FHS provide lower estimates because they reflect the lower volatility state at the end of the sample. Across all methods, INGA.AS is clearly the riskiest asset, while the loan return contributes almost no measurable market risk.

Effect of sample choice

The sample comparison using the normal method in Table 5 reveals very strong sensitivity to the estimation window. During the GFC crisis period alone ($n = 151$ observations), VaR estimates are roughly 2.5–3 times the full-sample figures: INGA.AS rises from 8.29% to 21.58%, ˆGSPC from 3.20% to 8.36%, and EURUSD=X from 2.18% to 5.83%. Excluding the GFC lowers the estimates substantially, confirming that the crisis is the dominant driver of the full-sample risk measures. This has a clear practical implication: using only calm-period data would seriously understate risk, while using only crisis data would overstate risk in normal times. The full sample therefore gives the most balanced estimate.

Table 5: Sample comparison — Normal VaR and ES (99%) for each individual asset across three estimation windows. All values are expressed as percentages.

	Full (2006–2013) $n = 2063$		**Non-crisis (excl. GFC)** $n = 1912$		**Crisis only (GFC)** $n = 151$	
Asset	**VaR**	**ES**	**VaR**	**ES**	**VaR**	**ES**
INGA.AS	8.2906	9.4936	6.2817	7.2041	21.5815	24.5672
SHEL	4.2212	4.8396	3.1956	3.6680	10.9170	12.4666
XOM	3.7933	4.3509	2.9006	3.3298	9.6271	11.0149
ˆAEX	3.4004	3.8950	2.6900	3.0860	8.3823	9.5413
ˆGSPC	3.1969	3.6652	2.4085	2.7658	8.3637	9.5361
EURUSD=X	2.1757	2.4935	1.5771	1.8086	5.8331	6.6729
log_loan_ret	0.0083	0.0109	0.0085	0.0112	0.0000	0.0000

Portfolio VaR and ES

Now the VaR and ES will be calculated using the same methods but for the whole portfolio. Each method will be evaluated, and the differences between them will again be analysed and compared.

The portfolio consists of all seven components: the three stocks (INGA.AS, SHEL, XOM), the two indices (ˆAEX, ˆGSPC), the FX rate (EURUSD=X), and the loan component. The weights sum to 100%, with the largest allocation to the loan (20%), followed by 15% in equities (except ING) and indices, and 10% in ING and EUR/USD. The portfolio has an average daily return of 0.0101% and a standard deviation of 1.1050%, which is much lower than the most volatile individual assets. This clearly illustrates the diversification effect: even though assets such as INGA.AS are quite risky, combining them with other assets and a near risk-free loan reduces total portfolio risk significantly.

Method 1 — Variance-Covariance (Normal)

Under the normal VCV method, the portfolio VaR is 2.56% and ES is 2.94%, as can be seen in Table 6. This is much lower than the standalone VaR of INGA.AS (8.29%) and also below the other individual assets from Task 3, confirming the diversification effect. Because correlations are far below one, as shown in the CCC correlation matrix in Table 7, and part of the portfolio is low-risk, overall volatility is reduced. The 20% weight in the near-riskless loan component also brings down volatility substantially.

Compared with the individual asset results, the normal portfolio VaR appears economically sensible. It captures average co-movement through the covariance matrix, so it reflects diversification directly rather than merely averaging the constituent VaRs. However, the normal model should still be seen as a lower bound, since it assumes thin tails and will underestimate extreme losses, especially in a sample that includes the GFC.

Table 6: Portfolio 1-day 99% VaR and ES under five methods, full sample 2006–2013. For the Student-t method, results are shown for df $\in \{3, 4, 5, 6\}$. All values are expressed as percentages.

	Method	**VaR (%)**	**ES (%)**
1.	VCV Normal	2.5606	2.9350
2.	VCV Student-t, df = 3	2.8868	4.4578
	VCV Student-t, df = 4	2.9177	4.0691
	VCV Student-t, df = 5	2.8701	3.8010
	VCV Student-t, df = 6	2.8254	3.6283
3.	Historical Simulation	3.5105	4.8896
4.	GARCH(1,1)-CCC	1.3985	1.6037
5.	FHS-EWMA	1.5280	1.8935

Table 7: CCC correlation matrix for the seven portfolio components, estimated over the full sample 2006–2013.

	INGA.AS	**SHEL**	**XOM**	**ˆAEX**	**ˆGSPC**	**EURUSD=X**	**log_loan**
INGA.AS	1.000	0.371	0.354	0.818	0.521	0.093	−0.001
SHEL	0.371	1.000	0.699	0.538	0.659	0.208	−0.004
XOM	0.354	0.699	1.000	0.467	0.728	0.127	0.001
ˆAEX	0.818	0.538	0.467	1.000	0.628	0.098	−0.019
ˆGSPC	0.521	0.659	0.728	0.628	1.000	0.100	−0.023
EURUSD=X	0.093	0.208	0.127	0.098	0.100	1.000	0.025
log_loan	−0.001	−0.004	0.001	−0.019	−0.023	0.025	1.000

Method 2 — Variance-Covariance (Student-t)

The QQ-plots select df = 3 for the portfolio, as can be seen in Figure 5, indicating strong fat tails. This is consistent with the individual asset results, which also preferred df = 3. Diversification reduces volatility but does not eliminate tail dependence or crisis-driven extreme losses.

Moving from normal to Student-t increases VaR from 2.56% to 2.89%, as can be seen in Table 6, which is a moderate increase; however, ES rises much more, from 2.94% to 4.46%. This shows that tail losses are significantly more severe than the normal model suggests. It also mirrors the individual-asset results from earlier: introducing heavy tails does not dramatically change the quantile itself, but it does change the expected severity of losses beyond that quantile. Even when diversification keeps the VaR threshold relatively low, the expected loss in the extreme left tail will still be much larger than a normal model would suggest.

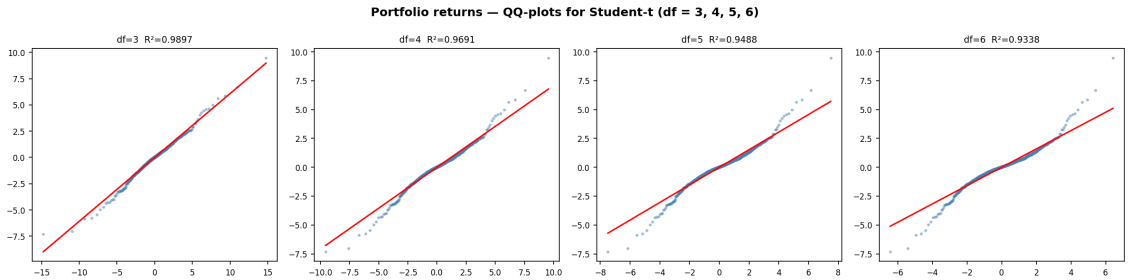

Figure 5: Normal QQ-plot of portfolio log returns over the full sample 2006–2013. Departures from the diagonal confirm fat tails at the portfolio level and support the selection of df = 3 for the Student-t VCV method.

Method 3 — Historical Simulation

Historical Simulation gives the highest stable estimates for the portfolio, as shown in Table 6: VaR of 3.51% and ES of 4.89%. These exceed both normal and Student-t results, meaning the empirical distribution contains more extreme losses than these parametric models capture. This is expected given the GFC in the sample. Compared to individual assets, risk for the whole portfolio is still lower due to diversification, but clearly higher than the normal estimate. The difference between 2.56% (normal) and 3.51% (Historical Simulation) is economically meaningful and shows that relying solely on covariance-based models can underestimate crisis risk.

Method 4 — GARCH(1,1)-CCC

The GARCH-CCC method produces the lowest estimates: VaR of 1.40% and ES of 1.60% for the whole portfolio. The reason is the same as in Task 3: it forecasts risk based on current market conditions at end-2013, a calm period, rather than the full crisis-inclusive sample. The rolling VaR plot in Figure 6 confirms this: risk peaks around 2009 and then declines towards 2013. The model is pricing the final segment of that trajectory and not the full path. Because of this, the GARCH estimate is not "too low"; it simply answers a different question. Historical Simulation asks: "What losses occurred over the whole crisis-inclusive history?" GARCH asks: "Given the volatility state at end-2013, what is tomorrow's likely downside?"

The correlation matrix in Table 7 supports the diversification story. Equity assets are positively correlated — most notably INGA.AS with ˆAEX (0.818) — but not perfectly so. EURUSD=X and especially the loan show very low correlations with the equity components, which helps reduce overall portfolio risk. This relatively low-correlation structure is what allows the portfolio VaR to remain well below the VaR of the most volatile constituents.

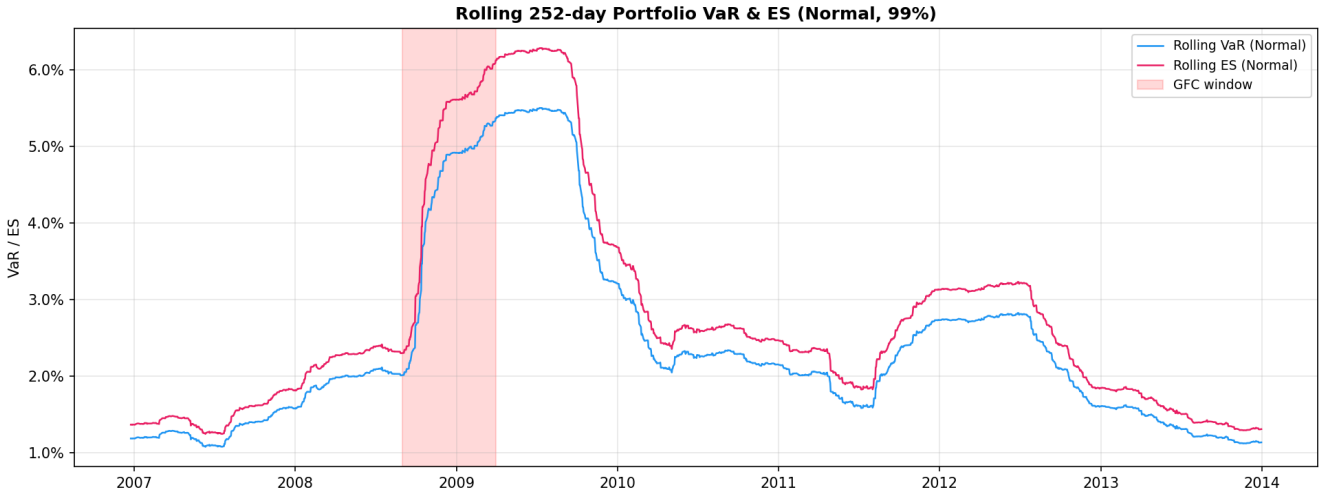

Figure 6: Rolling 252-day portfolio VaR and ES computed using the normal method. The sharp increase around the 2008–2009 GFC window and the subsequent gradual decline towards end-2013 illustrate the strong sensitivity of risk estimates to the estimation window.

Method 5 — Filtered Historical Simulation (EWMA)

For the full portfolio, FHS-EWMA produces a VaR of 1.53% and an ES of 1.89%, as shown in Table 6. This is lower than the FHS estimates of the individual assets: for example, INGA.AS has a VaR of 4.20% and ES of 5.49%, while even more stable assets such as ˆGSPC (VaR 1.73%) and EURUSD=X (VaR 0.83%) are comparable to or above the total portfolio risk. This again clearly reflects diversification: losses do not occur simultaneously across all assets. The reduction in VaR is largest for volatile components such as INGA.AS and XOM, whose individual tail risk is much higher, showing that their risks are not perfectly correlated with the rest of the portfolio. At the same time, the portfolio VaR remains meaningful, confirming that broad market shocks (especially during the GFC) are still captured.

The position of FHS between GARCH and Historical Simulation is the same as for the individual assets. The EWMA filter captures time-varying volatility and scales past crisis observations downward when current volatility is low. As a result, FHS remains above GARCH (since it still uses the empirical residual distribution) but below Historical Simulation, which assigns equal weight to all observations.

The crisis-only FHS result (VaR = 14.78%, ES $\approx$ 1956.87%) shown in Table 8 is not economically plausible. These values are far outside any realistic range for daily losses and are most likely caused by numerical instability. In a short crisis sample, EWMA scaling can produce extremely large standardised residuals, which then explode when aggregated at the portfolio level. This result should therefore be treated as unstable and not used for interpretation or comparison.

Effect of sample choice

The effect of sample choice is very large at the portfolio level, just as it was for the individual assets, as can be seen in Table 8. Under the normal method, portfolio VaR rises from 1.90% in the non-crisis sample to 6.82% in the crisis-only sample. Student-t rises from 2.14% to 7.65%, and Historical Simulation from 2.34% to 7.11%. These are very large differences, showing that the portfolio's estimated risk is dominated by whether the GFC is included in the estimation window. The same conclusion follows from the rolling 252-day portfolio VaR and ES shown in Figure 6.

Table 8: Portfolio 1-day 99% VaR and ES under five methods across three estimation windows. The crisis-only FHS-EWMA result is flagged as numerically unstable (see text). All values are expressed as percentages.

		Full (2006–2013)		**Crisis only (GFC)**	
	Method	**VaR (%)**	**ES (%)**	**VaR (%)**	**ES (%)**
1.	VCV Normal	2.5606	2.9350	6.8226	7.7753
2.	VCV Student-t, df = 3	2.8868	4.4578	7.6526	11.6497
3.	Historical Simulation	3.5105	4.8896	7.1146	7.9103
4.	GARCH(1,1)-CCC	1.3985	1.6037	5.7148	6.5062
5.	FHS-EWMA	1.5280	1.8935	14.7777	1956.87*

* Numerically unstable; see main text for discussion.

Conclusion

The portfolio-level results show clearly that diversification reduces risk but does not eliminate fat-tail exposure. Even though the portfolio standard deviation is much lower than the volatility of INGA.AS or SHEL individually, the Student-t QQ-plots still select df = 3 and both ES and HS VaR remain meaningfully above the normal benchmark. All methods tell approximately the same story: the portfolio is notably safer than its riskiest asset, but its downside risk is still very sensitive to crisis periods and to the chosen modelling framework.

Backtesting of VaR and ES

In this subsection, the five portfolio risk models from the previous section are backtested: (i) Variance–Covariance (VCV) Normal, (ii) VCV Student-t, (iii) Historical Simulation (HS), (iv) GARCH(1,1)-CCC, and (v) Filtered Historical Simulation with EWMA (FHS-EWMA). The assignment requires comparing the *number of VaR violations per year* to the *expected number of violations*, comparing the *average ES per year* to the *actual average shortfall*, and investigating the *dependence in the violations graphically*. We therefore evaluate both the *frequency* and the *severity* of tail losses.

Backtesting setup

Backtesting is performed on the *portfolio* rather than on the individual assets. Consistent with the earlier tasks, daily *log returns* are used throughout. Let $r_{p,t}$ denote the portfolio log return on day t. The realized loss is then defined as

$$L_t = -r_{p,t}.$$

A VaR violation occurs whenever

$$L_t > VaR_t.$$

For the daily-grid methods (Normal, Student-t, HS, and FHS-EWMA), one-day-ahead forecasts are generated using a *rolling 252-day estimation window*. Hence, for each day t, the model is estimated using only the previous 252 trading days and is then compared with the realized portfolio loss on day t. This creates a fully out-of-sample backtest.

For GARCH(1,1)-CCC, rolling estimation over every single day was computationally very expensive. Therefore, the GARCH-CCC backtest was performed on a *reduced grid*, with a one-day-ahead forecast computed every fifth backtest date. The expected number of violations for GARCH-CCC is therefore computed using the corresponding reduced number of forecast dates. This reduced-grid setup is explicitly indicated whenever GARCH-CCC is discussed.

The confidence level is 99%, so the expected violation probability is 1% per forecast day.

Backtesting of VaR: frequency of violations

Table 9 reports the overall number of VaR violations for each method over the full backtest period. The daily-grid methods are directly comparable; GARCH-CCC is reported separately on its reduced grid.

Table 9: Overall VaR backtesting summary.

Method	Valid forecasts	Actual violations	Expected violations	Violation rate
VCV Normal	1811	44	18.11	2.43%
VCV Student-t	1811	33	18.11	1.82%
Historical Simulation	1811	27	18.11	1.49%
FHS-EWMA	1787	17	17.87	0.95%
GARCH(1,1)-CCC (reduced grid)	363	12	3.63	3.31%

Several conclusions follow immediately from Table 9. First, the Normal model performs worst, with a violation rate of 2.43%, which is far above the target 1%. This indicates substantial underestimation of tail risk. Second, Student-t improves materially on the Normal model, which is consistent with the lecture material: heavier-tailed innovations provide a better description of financial returns than the Gaussian benchmark. Third, Historical Simulation improves further and is more conservative than both parametric VCV methods. Finally, FHS-EWMA is the model closest to the target violation rate, with 0.95%, suggesting that time-varying volatility is important for the portfolio under study.

The reduced-grid GARCH-CCC backtest still produces too many violations (3.31%), indicating that the model remains too optimistic overall, despite its ability to react to changing volatility. This suggests that, in this portfolio, CCC-GARCH is not sufficient to fully capture the dynamics of tail risk.

To see how these results vary over time, Table 10 reports the yearly number of actual and expected violations for the daily-grid methods. Table 11 reports the corresponding reduced-grid results for GARCH-CCC.

Table 10: Yearly VaR backtesting results for the daily-grid methods. Entries are reported as actual / expected violations.

Year	VCV Normal	VCV Student-t	Historical Simulation	FHS-EWMA
2006	0 / 0.04	0 / 0.04	0 / 0.04	0 / 0.04
2007	10 / 2.58	8 / 2.58	7 / 2.58	4 / 2.56
2008	17 / 2.54	14 / 2.54	12 / 2.54	2 / 2.49
2009	0 / 2.60	0 / 2.60	0 / 2.60	2 / 2.54
2010	5 / 2.60	2 / 2.60	0 / 2.60	2 / 2.56
2011	11 / 2.59	8 / 2.59	7 / 2.59	3 / 2.56
2012	0 / 2.58	0 / 2.58	0 / 2.58	2 / 2.56
2013	1 / 2.58	1 / 2.58	1 / 2.58	2 / 2.56

Table 11: Yearly VaR backtesting results for GARCH(1,1)-CCC on the reduced grid. Entries are reported as actual expected violations.

Year	GARCH(1,1)-CCC (actual / expected)	Reduced-grid violation rate
2006	0 / 0.01	0.00%
2007	4 / 0.52	7.69%
2008	0 / 0.51	0.00%
2009	2 / 0.52	3.85%
2010	2 / 0.52	3.85%
2011	2 / 0.51	3.92%
2012	2 / 0.52	3.85%
2013	0 / 0.52	0.00%

The yearly results highlight a clear regime-dependent pattern. In 2007, 2008, and 2011, all daily-grid models produce too many violations, especially the Normal model. This indicates that the models underestimated risk in turbulent periods. By contrast, in 2009 and 2012 there are no VaR violations for Normal, Student-t, and HS, implying that these models became very conservative after the crisis shocks had entered the rolling estimation windows. This pattern — too many violations during stress, then too few afterwards — is typical of models that react to large shocks with a lag.

Among the daily-grid methods, FHS-EWMA is closest to the desired violation frequency over the full sample. In particular, FHS-EWMA reacts strongly during and after the global financial crisis and therefore avoids the very large number of violations observed under the static Normal specification.

Figure 7 shows the realized portfolio loss together with the rolling 99% VaR forecasts for the fast daily-grid methods (Normal, Student-t, and HS). Figure 8 shows the corresponding FHS-EWMA VaR forecast, which reacts more strongly to the crisis episode. Figure 9 reports the reduced-grid GARCH-CCC VaR path.

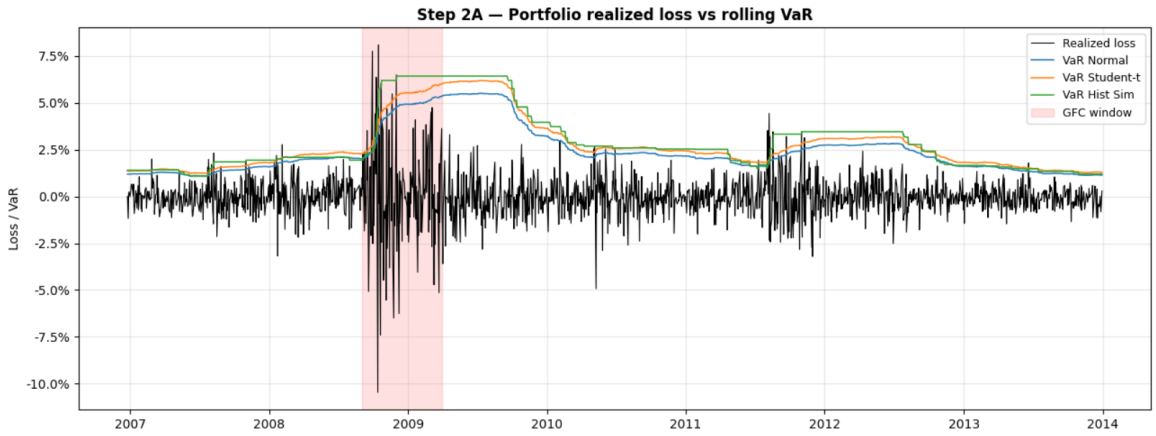

Figure 7: Realized portfolio loss versus rolling 99% VaR forecasts for the daily-grid methods: VCV Normal, VCV Student-t, and Historical Simulation. The shaded area indicates the global financial crisis window.

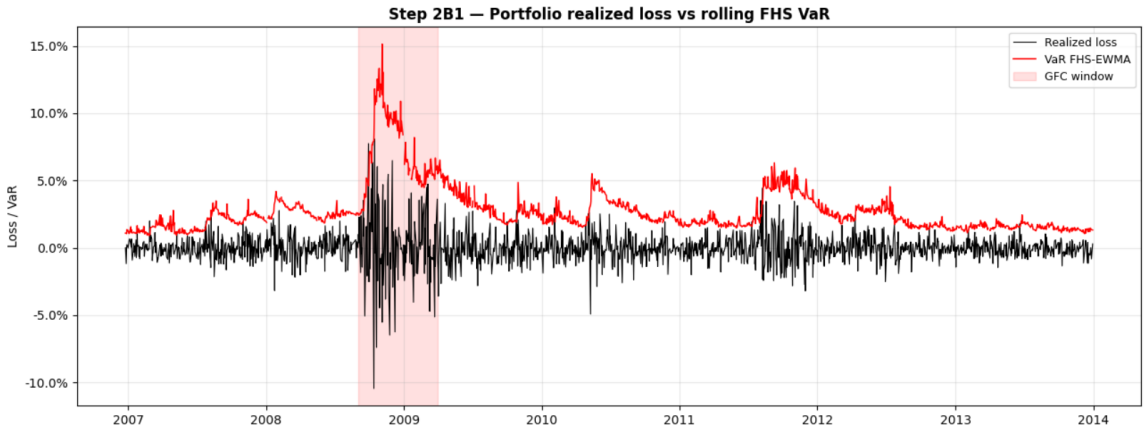

Figure 8: Realized portfolio loss versus rolling 99% VaR forecast under FHS-EWMA. The FHS specification responds strongly to changes in volatility and yields a violation rate closest to the 1% target.

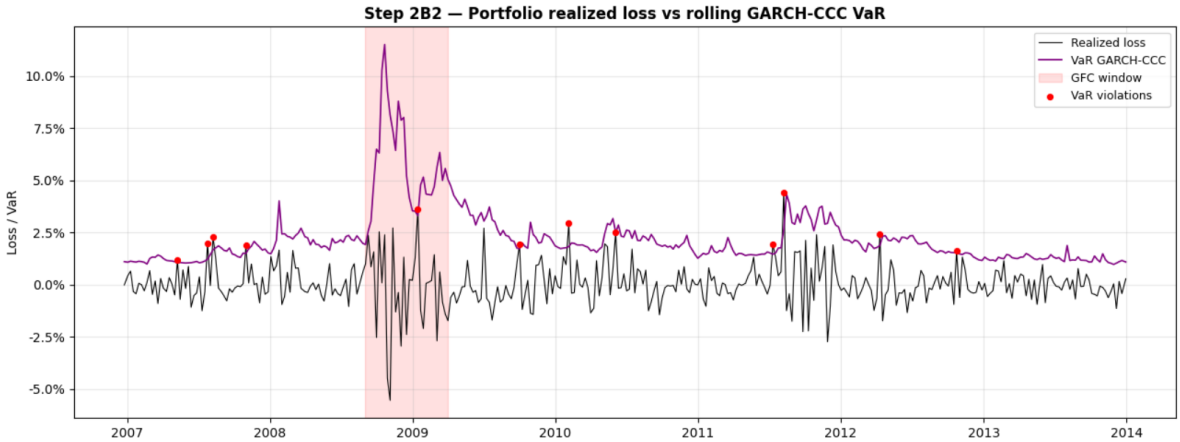

Figure 9: Realized portfolio loss versus rolling 99% VaR forecast under GARCH(1,1)-CCC, evaluated on a reduced grid (every fifth backtest date).

Backtesting of ES: predicted ES versus actual average shortfall

The assignment also requires comparing the *average ES per year* to the *actual average shortfall*. For each method and year, we therefore consider only the days on which that model produced a VaR violation. On those violation days, we compute:

- the average predicted ES of the model;
- the actual average shortfall, i.e. the average realized loss conditional on a VaR violation.

A positive difference

$$\text{actual average shortfall} - \text{predicted ES} > 0$$

means that the model underestimates the severity of tail losses. A negative difference means that the model is conservative in the tail.

Table 12 reports the yearly ES backtesting results for the daily-grid methods. Table 13 reports the reduced-grid GARCH-CCC results.

Table 12: Yearly ES backtesting results for the daily-grid methods. Entries are reported as predicted ES / actual average shortfall (in %). A dash indicates that no VaR violations occurred in that year for that model.

Year	VCV Normal	VCV Student-t	Historical Simulation	FHS-EWMA
2006	–	–	–	–
2007	1.43 / 1.72	2.16 / 1.81	1.77 / 1.84	13.50 / 1.85
2008	3.33 / 4.49	4.74 / 4.62	4.30 / 4.78	4.12 / 5.63
2009	–	–	–	1.89 / 2.36
2010	2.67 / 2.63	3.65 / 2.58	–	3.17 / 2.13
2011	2.26 / 2.93	3.48 / 3.29	2.66 / 3.08	4.33 / 3.29
2012	–	–	–	2.10 / 2.02
2013	1.52 / 1.79	2.33 / 1.79	1.56 / 1.79	1.66 / 1.56

Table 13: Yearly ES backtesting results for GARCH(1,1)-CCC on the reduced grid. Entries are reported as predicted ES / actual average shortfall (in %).

Year	GARCH(1,1)-CCC	Comment
2006	–	no violations
2007	1.57 / 1.84	slight underestimation
2008	–	no violations
2009	2.88 / 2.79	close fit
2010	2.50 / 2.74	slight underestimation
2011	2.37 / 3.18	underestimation
2012	1.97 / 2.02	close fit
2013	–	no violations

The ES backtest refines the conclusions from the VaR analysis. The Normal model systematically underestimates tail severity in the years with many violations: for example, in 2008 its average predicted ES is 3.33%, whereas the actual average shortfall equals 4.49%. The same pattern appears in 2007 and 2011. Hence, the Gaussian model performs poorly not only in terms of the *frequency* of violations, but also in terms of the *severity* of losses once the VaR threshold is breached.

Student-t performs much better. In most years, the predicted ES is close to or slightly above the realized average shortfall, indicating a more realistic treatment of heavy tails. Historical Simulation also performs well and is particularly accurate in 2007, 2008, 2011, and 2013. In these years, the gap between the predicted ES and the actual average shortfall is small relative to the Normal model.

FHS-EWMA gives the best calibration for VaR *frequency*, but its ES results are more mixed. In 2007, predicted ES is extremely high relative to the realized average shortfall (13.50% versus 1.85%), implying strong overconservatism. In 2008, however, FHS-EWMA underestimates tail severity (4.12% versus 5.63%). Hence, FHS-EWMA performs very well as a VaR model, but not as consistently as an ES model. This is an important finding: a model that gets the *number* of VaR violations right does not automatically describe the *magnitude* of extreme losses correctly.

GARCH(1,1)-CCC shows a mixed picture. In some years (2009, 2012), the predicted ES is very close to the actual average shortfall, but in others (2007, 2011) the model still underestimates tail severity. Together with the reduced-grid VaR results, this suggests that GARCH-CCC is reactive, but not clearly superior to the simpler alternatives for this portfolio.

Figure 10 summarizes the yearly ES comparison graphically.

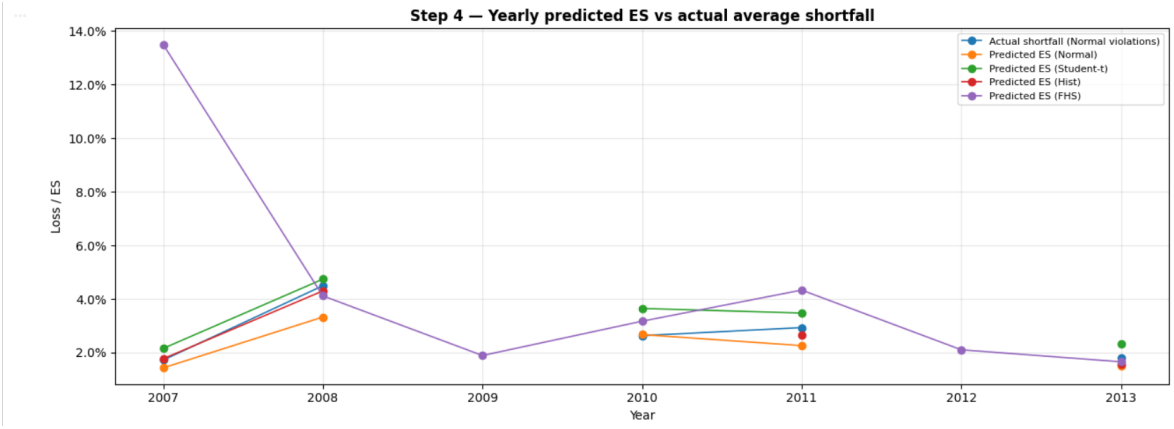

Figure 10: Yearly predicted ES compared with the actual average shortfall on violation days.

Dependence and clustering of VaR violations

A correct VaR model should not only have approximately the correct number of violations, but the violations should also occur in a roughly random manner rather than in clear clusters. To assess this, we inspect the time series of violation indicators and the autocorrelation functions (ACFs) of the indicators for the daily-grid methods.

Figure 11 shows that violations are concentrated around the global financial crisis, rather than being evenly spread across time. This indicates dependence in the violation process. The ACFs also show small but positive serial dependence at short lags, especially for the Normal, Student-t, and Historical Simulation models. Thus, the Bernoulli independence assumption is not fully satisfied.

To summarize the degree of clustering, Table 14 reports the total number of violations and the number of separate violation clusters. If the number of clusters is close to the number of violations, most exceedances are isolated; if the number of clusters is much smaller, exceedances are grouped in bursts.

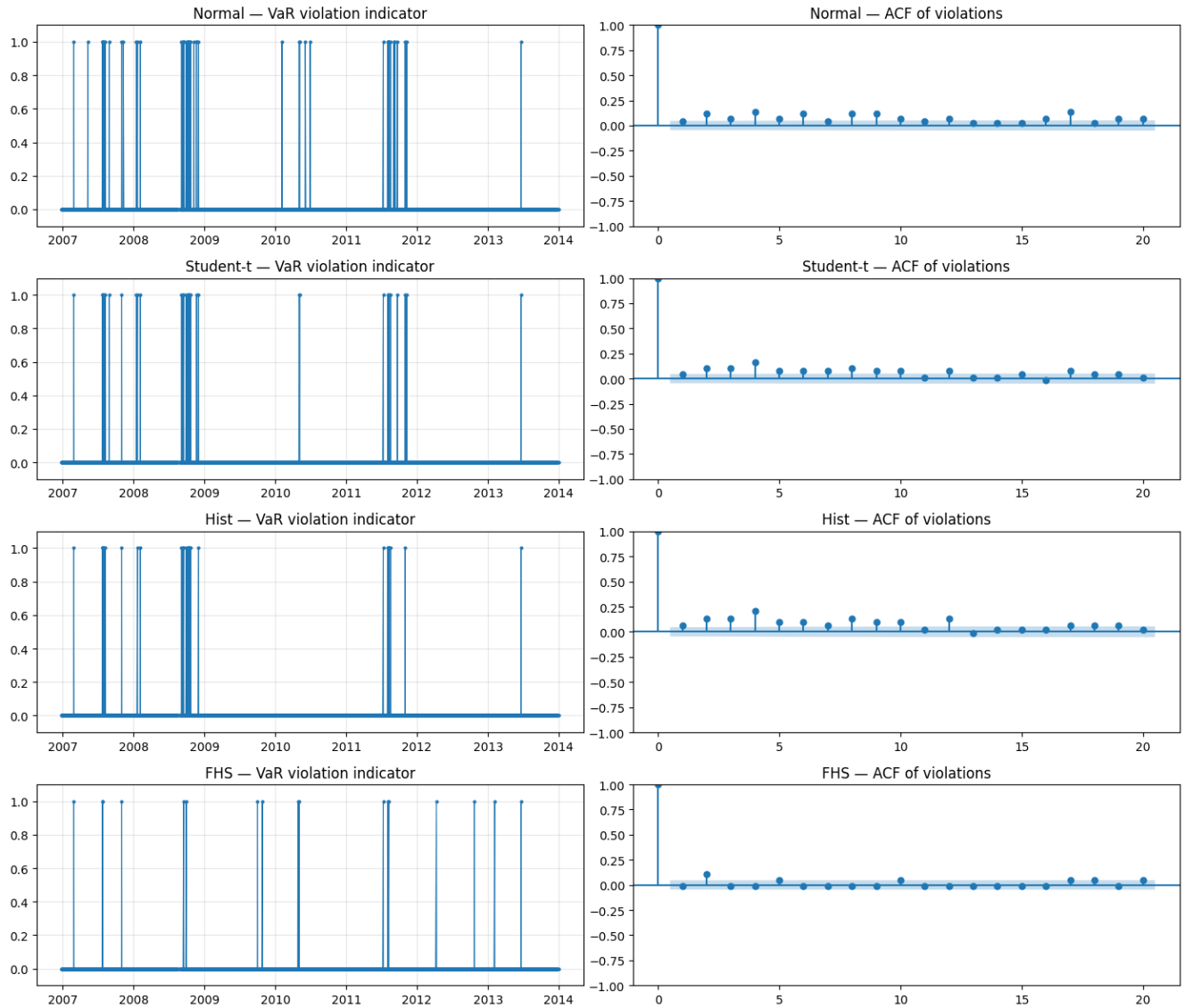

Figure 11: Violation indicators and autocorrelation functions for the daily-grid methods. Clustering is most visible around the global financial crisis.

Table 14: Violation clustering summary for the daily-grid methods.

Method	Total violations	Number of clusters
VCV Normal	44	41
VCV Student-t	33	31
Historical Simulation	27	25
FHS-EWMA	17	17

The clustering results again support the previous conclusions. FHS-EWMA is the strongest VaR model in terms of violation dependence: the number of clusters equals the number of violations, meaning that violations are almost entirely isolated. By contrast, the other models exhibit more grouping of violations, especially around the crisis period. Hence, the evidence suggests that FHS-EWMA comes closest to satisfying both the *frequency* and the *independence* requirements for VaR backtesting.

Overall conclusion and recommendation

The backtesting results show that the choice of model materially affects the quality of risk forecasts. The main findings are as follows:

- The **VCV Normal** model performs worst. It generates far too many violations in stress periods and systematically underestimates the severity of losses on violation days.
- The **VCV Student-t** model is a substantial improvement over the Normal model. It reduces the number of violations and provides a better description of tail severity.
- **Historical Simulation** performs well overall and provides realistic ES estimates, but still exhibits some clustering in the violations.
- **FHS-EWMA** performs best for VaR backtesting: its overall violation frequency is closest to the 1% target, and its violations are the least clustered. However, its ES performance is less stable, since it can be overly conservative in some years and underestimate tail severity in others.
- **GARCH(1,1)-CCC** is computationally demanding and, in this application, does not clearly dominate the simpler alternatives. On the reduced grid it still produces too many violations and only mixed ES results.

Taken together, these findings suggest the following recommendation for this portfolio. If the main goal is **accurate VaR forecasting**, **FHS-EWMA** is the preferred method, because it achieves the best overall calibration and the cleanest violation pattern. If the main goal is a more balanced description of both **VaR and ES**, **Historical Simulation** and **VCV Student-t** provide the strongest overall compromise. By contrast, the Gaussian VCV model should not be preferred for this portfolio, since it performs weakest both in terms of the number of violations and in terms of the size of losses once violations occur.

Multi-Day Value-at-Risk Analysis

This analysis addresses the estimation of 5-day and 10-day VaR for the portfolio using the Historical Method on non-overlapping data. The question is whether the square-root-of-time rule provides a reasonable approximation of multi-day VaR for this portfolio. To answer this, we compare the scaled VaR predictions against the historic VaR estimates, evaluating the empirical violation rates at three confidence levels: $\alpha \in \{99\%, 97.5\%, 95\%\}$.

Non-Overlapping h-Day Return Construction

Let the daily portfolio log-returns be denoted by $\{r_t\}_{t=1}^{T}$, where $T = 2,063$ trading days spanning approximately seven years from 2006 to 2013. The daily portfolio return is computed as the weighted sum of individual asset returns:

$$r_t = \sum_{i=1}^{d} w_i r_{i,t} = \mathbf{w}'\mathbf{r}_t \tag{2}$$

where $\mathbf{w} = (w_1, \ldots, w_d)'$ is the vector of fixed portfolio weights and $\mathbf{r}_t$ is the vector of individual asset log-returns on day t.

For a given horizon $h \in \{1, 5, 10\}$, we partition the sequence of daily returns into n_h non-overlapping blocks of length h:

$$n_h = \left\lfloor \frac{T}{h} \right\rfloor \tag{3}$$

where $\lfloor \cdot \rfloor$ denotes the floor function. For our sample:

$$\begin{aligned} n_1 &= \lfloor 2063/1 \rfloor = 2063 \quad \text{(all daily observations)} \\ n_5 &= \lfloor 2063/5 \rfloor = 412 \quad \text{(complete 5-day blocks)} \\ n_{10} &= \lfloor 2063/10 \rfloor = 206 \quad \text{(complete 10-day blocks)} \end{aligned}$$

The h-day log-return for block j is obtained by summing the h consecutive daily log-returns within that block:

$$R_j^{(h)} = \sum_{k=1}^{h} r_{(j-1)h+k}, \quad j = 1, 2, \ldots, n_h \tag{4}$$

The use of non-overlapping data is crucial as overlapping h-day returns (e.g., computing a 5-day return every day) would introduce serial correlation, violating the independence assumption required for reliable VaR estimation via Historical Simulation.

Historical Simulation VaR

For a given return series $\{R_i\}_{i=1}^{n}$, the Historical Simulation VaR at confidence level α is defined as the negative of the α-quantile of the empirical return distribution. Let $R_{(1)} \leq R_{(2)} \leq \cdots \leq R_{(n)}$ denote the ordered returns. Then:

$$\widehat{\mathrm{VaR}}_{\alpha}^{(h)} = -\hat{F}_{R^{(h)}}^{-1}(\alpha) = -R_{(k)} \tag{5}$$

where the index k is determined by:

$$k = \lceil n \cdot \alpha \rceil \tag{6}$$

with $\lceil \cdot \rceil$ denoting the ceiling function. This ensures that approximately $\alpha \cdot 100\%$ of the historical losses fall below $\widehat{\mathrm{VaR}}_{\alpha}^{(h)}$. This method is fully non-parametric, making no distributional assumptions about the return generating process. It captures the empirical tail behavior directly from historical data, including any skewness, heavy tails, or other non-normal characteristics present in the sample.

Square-Root-of-Time Rule

Under the classical assumptions that daily returns are:

1. Independently and identically distributed (i.i.d.)
2. Normally distributed: $r_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$
3. With negligible mean: $\mu \approx 0$

the variance of h-day returns scales linearly with the horizon:

$$\mathrm{Var}\left(R^{(h)}\right) = \mathrm{Var}\left(\sum_{k=1}^{h} r_{t+k}\right) = \sum_{k=1}^{h} \mathrm{Var}(r_{t+k}) = h \cdot \sigma^2 \tag{7}$$

Consequently, the standard deviation scales with the square root of time: $\sigma_h = \sqrt{h} \cdot \sigma_1$. Since VaR under normality is proportional to the standard deviation ($\mathrm{VaR}_\alpha = z_\alpha \cdot \sigma$), the h-day VaR can be predicted by scaling the 1-day VaR:

$$\mathrm{VaR}_\alpha^{(h,\mathrm{sqrt})} = \sqrt{h} \cdot \mathrm{VaR}_\alpha^{(1)} \tag{8}$$

For our analysis, we compute:

$$\mathrm{VaR}_\alpha^{(5)} = \sqrt{5} \cdot \mathrm{VaR}_\alpha^{(1)} \tag{9}$$

$$\mathrm{VaR}_\alpha^{(10)} = \sqrt{10} \cdot \mathrm{VaR}_\alpha^{(1)} \tag{10}$$

Comparison Framework

To assess the validity of the square-root rule, we compare:

- **Historical Non-Overlapping h-Day VaR**: $\mathrm{VaR}_\alpha^{(h,\mathrm{H})}$ computed directly from non-overlapping h-day returns
- **Predicted VaR**: $\mathrm{VaR}_\alpha^{(h,\mathrm{sqrt})} = \sqrt{h} \cdot \mathrm{VaR}_\alpha^{(1,\mathrm{HS})}$

We evaluate each VaR estimate by computing its empirical violation rate. For a given VaR estimate $\widehat{\mathrm{VaR}}_\alpha^{(h)}$, the h-day loss for block j is:

$$L_j^{(h)} = -R_j^{(h)} \tag{11}$$

A violation occurs when the realized loss exceeds the VaR forecast:

$$I_j = \mathbf{1}\left\{L_j^{(h)} > \widehat{\mathrm{VaR}}_\alpha^{(h)}\right\} \tag{12}$$

Under a correctly specified model at confidence level α, the violation probability is $p = 1 - \alpha$. Thus, $I_j \overset{\text{i.i.d.}}{\sim} \mathrm{Bernoulli}(1-\alpha)$, and the empirical violation rate $\hat{p} = \frac{1}{n_h}\sum_{j=1}^{n_h} I_j$ should approximate $1 - \alpha$.

Binomial Test for Unconditional Coverage Let $I_{\text{total}} = \sum_{j=1}^{n_h} I_j$ be the total number of violations. Under H_0 of a correctly specified model, $I_{\text{total}} \sim \mathrm{Binomial}(n_h, 1-\alpha)$ with $\mathbb{E}[I_{\text{total}}] = n_h(1-\alpha)$. We test H_0 against the two-sided alternative $H_1 : I_{\text{total}} \neq n_h(1-\alpha)$ at the 5% significance level. The p-value is computed as $p = 2 \cdot \min\{P(X \leq I_{\text{total}}), P(X \geq I_{\text{total}})\}$ with $X \sim \mathrm{Binomial}(n_h, 1-\alpha)$. For large samples where $n_h(1-\alpha) \geq 10$ and $n_h\alpha \geq 10$, the normal approximation applies:

$$I_{\text{total}} \overset{\text{approx}}{\sim} \mathcal{N}\left(n_h(1-\alpha),\ n_h\alpha(1-\alpha)\right) \tag{13}$$

A small p-value (< 0.05) indicates that the VaR model is miscalibrated.

Independence Test via Spacings Beyond the correct number of violations, a good VaR model should produce violations that occur independently. Let $T_1, \ldots, T_I$ denote the violation block indices, with $T_0 \equiv 0$. The spacings $S_j = T_j - T_{j-1}$ represent the waiting times between consecutive violations. Under H_0, these spacings follow a geometric distribution with success probability $p = 1 - \alpha$:

$$S_j \overset{\text{i.i.d.}}{\sim} \text{Geometric}(p = 1 - \alpha), \quad \mathbb{E}[S_j] = \frac{1}{1 - \alpha} \tag{14}$$

For small $1 - \alpha$ (e.g., 0.01, 0.025, 0.05), the geometric distribution is well-approximated by an exponential distribution: $S_j \overset{\text{approx}}{\sim} \text{Exponential}(\lambda = 1 - \alpha)$. We apply the Kolmogorov-Smirnov (KS) test to compare the empirical spacings against this exponential distribution. A p-value below 0.05 indicates significant clustering of violations, invalidating the independence assumption.

Results

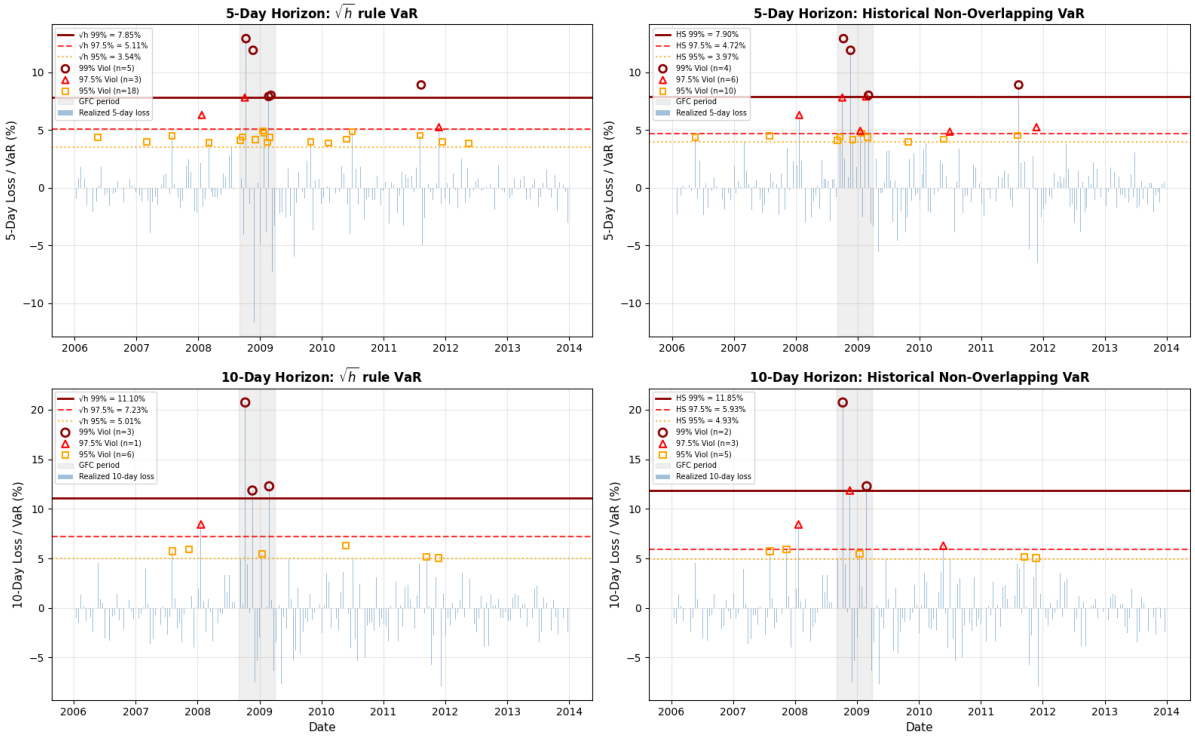

Figure 12: Multi-day VaR backtest: Realized losses versus static VaR estimates for 5-day (top row) and 10-day (bottom row) horizons. Left panels show $\sqrt{h}$ rule VaR; right panels show Historical Non-Overlapping h-Day VaR computed directly from non-overlapping returns. Horizontal lines indicate VaR thresholds at 99% (solid), 97.5% (dashed), and 95% (dotted) confidence levels. Violations are marked by circles (99%), triangles (97.5%), and squares (95%). Gray shading denotes the Global Financial Crisis (Sep 2008–Mar 2009).

Figure 12 presents the realized h-day losses alongside the VaR estimates from both the $\sqrt{h}$ rule (left panels) and the Historical Non-Overlapping h-Day VaR (right panels). The horizontal lines indicate the full-sample VaR thresholds at 99% (solid), 97.5% (dashed), and 95% (dotted) confidence levels. Violations are marked hierarchically: dark red circles for 99% violations, red triangles for additional 97.5% violations, and orange squares for additional 95% violations. The gray shaded region denotes the Global Financial Crisis (GFC) period from September 2008 to March 2009.

Two key observations emerge from the visualization. First, violations cluster predominantly during the GFC period, reflecting the extreme market volatility experienced during the financial crisis and highlighting the limitation of constant full-sample VaR estimates that cannot adapt to changing market conditions. Second, the Historical Non-Overlapping h-Day VaR estimates (right panels) align closely with the empirical tail of the loss distribution, while the $\sqrt{h}$ rule VaR (left panels) shows some misalignment, particularly at the 97.5% confidence level where the thresholds appear overly conservative relative to the empirical violations.

Table 15 reports the full-sample Historical Simulation VaR estimates for all horizons and confidence levels.

Table 15: Full-Sample Historical Simulation VaR

Horizon	99% VaR	97.5% VaR	95% VaR
1-day	3.51%	2.29%	1.58%
5-day	7.90%	4.72%	3.97%
10-day	11.85%	5.93%	4.93%

Violation Analysis and Binomial Test

Table 16 summarizes the violation rates for both VaR estimation methods. The Historical Non-Overlapping h-Day VaR, computed directly from non-overlapping h-day returns, produces violation rates that are consistently close to their theoretical expectations. At the 99% confidence level, the 5-day and 10-day violation rates are 0.97% (versus expected 1.00%). At 97.5%, the rates are 2.43% (versus expected 2.50%). At 95%, the rates are 4.85% (versus expected 5.00%). This close alignment confirms that Historical Simulation on properly constructed non-overlapping data yields well-calibrated VaR estimates.

Table 16: Violation Rates by Method and Confidence Level

Confidence	Theoretical Rate	Historical Non-Overlapping h-Day VaR				$\sqrt{h}$ Rule VaR			
		5-day		10-day		5-day		10-day	
		Viol	Rate	Viol	Rate	Viol	Rate	Viol	Rate
99%	1.00%	4/412	0.97%	2/206	0.97%	5/412	1.21%	3/206	1.46%
97.5%	2.50%	10/412	2.43%	5/206	2.43%	8/412	1.94%	4/206	1.94%
95%	5.00%	20/412	4.85%	10/206	4.85%	26/412	6.31%	10/206	4.85%

For the $\sqrt{h}$ rule VaR, the violation rates show moderate deviations from the theoretical values. At 99% confidence, the rates are slightly elevated (1.21% for 5-day, 1.46% for 10-day), suggesting a mild underestimation of extreme tail risk. At 97.5%, the rates are consistently low (1.94% for both horizons), indicating that the scaled VaR is overly conservative at this confidence level. At 95%, the 5-day rate is elevated (6.31%) while the 10-day rate is close to expected (4.85%).

Table 17 presents the formal binomial test results for the $\sqrt{h}$ rule VaR. At all three confidence levels and both horizons, the binomial test fails to reject the null hypothesis of correct unconditional coverage at the 5% significance level (all p-values > 0.05).

It is important to note when the normal approximation to the binomial distribution is valid. The approximation requires both the expected number of violations and non-violations to be at least 10: $n_h(1-\alpha) \geq 10$ and $n_h\alpha \geq 10$. In our analysis, this condition holds only for the 97.5% VaR at the 5-day horizon, and for the 95% VaR at both 5-day and 10-day horizons. For the 99% VaR (both horizons) and the 97.5% VaR (10-day horizon), the expected number of violations is less than 10, so we must rely on the exact binomial test. Where valid, the normal approximation yields z-statistics of -0.726 ($p = 0.468$) for the 97.5% 5-day VaR, 1.221 ($p = 0.222$) for the 95% 5-day VaR, and -0.096 ($p = 0.924$) for the 95% 10-day VaR—all confirming the exact binomial test conclusions.

Table 17: Binomial Test Results for $\sqrt{h}$ Rule VaR

Confidence	Horizon	n	Violations	Rate	p-value
99%	5-day	412	5	1.21%	0.789
	10-day	206	3	1.46%	0.679
97.5%	5-day	412	8	1.94%	0.594
	10-day	206	4	1.94%	0.824
95%	5-day	412	26	6.31%	0.270
	10-day	206	10	4.85%	1.000

Independence of Violations

The spacings-based KS test evaluates whether violations occur independently rather than in clusters. Table 18 reports the results. At the 99% and 97.5% confidence levels, the KS test fails to reject the null hypothesis of independent violations for both horizons (all p-values > 0.05). The mean spacings are reasonably close to their theoretical expectations, although the small number of violations limits the power of the test.

At the 95% confidence level, the 5-day horizon exhibits a significant deviation from independence ($p = 0.008$), with a mean spacing of 12.7 periods compared to the expected 20.0 periods. This indicates that violations are occurring more frequently than expected under independence, consistent with the elevated violation rate of 6.31% observed at this confidence level. The 10-day horizon, however, shows no significant violation clustering ($p = 0.255$).

Table 18: Independence Test Results for $\sqrt{h}$ Rule VaR

Confidence	Horizon	Violations	Mean Spacing	Expected Spacing	KS p-value
99%	5-day	5	57.8	100.0	0.144
	10-day	3	27.0	100.0	0.145
97.5%	5-day	8	38.0	40.0	0.343
	10-day	4	20.2	40.0	0.477
95%	5-day	26	12.7	20.0	0.008
	10-day	10	15.2	20.0	0.255

Stress Testing

To examine the versatility and robustness of our portfolio, we applied shocks individually to each risk factor class: equities, FX, and interest rates. This stress testing exercise evaluates our portfolio under extreme but plausible market scenarios. We consider two complementary approaches: deterministic stress testing and distribution-based stress testing.

- **Deterministic Stress Testing:** Here we evaluate the direct profit and loss (P&L) effects resulting from instantaneous shocks to the components of our portfolio: equities, exchange rates, and interest rates.
- **Distribution-Based Stress Testing:** Using the Filtered Historical Simulation (FHS) framework, we adjust the return series to assess how the full loss distribution and VaR evolve under stress scenarios.

Deterministic Stress Testing Results

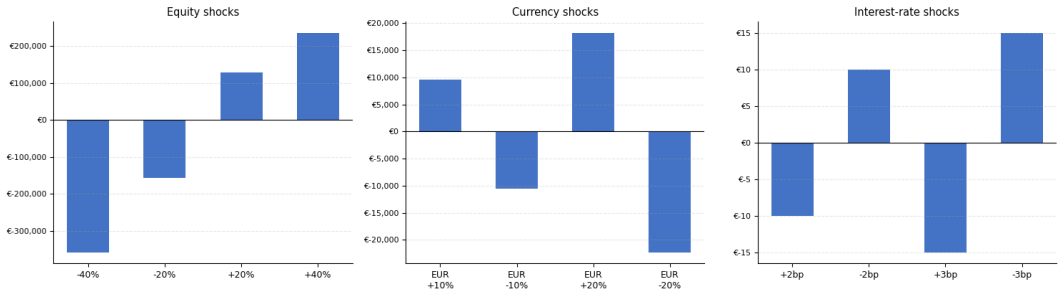

Figure 13: Deterministic stress test results: P&L impact from shocks to equities, currencies, and interest rates.

Figure 13 displays the deterministic effects on the P&L assuming a portfolio value of €1 million. Equity shocks have the largest impact, causing potential profits or losses of up to approximately €300,000. This is expected given that equities constitute the majority of our portfolio exposure. Currency shocks have the second largest effect, with approximately €22,000 impact for a 20% shock and €11,000 for a 10% shock. Interest rate shocks have negligible effects relative to the other risk factors, causing P&L changes of only €10 to €15.

Table 19: Portfolio Log Returns Under Equity Shocks

Shock Scenario	**Log Return (%)**
Equities (+20%)	+12.76
Equities (-20%)	-15.62
Equities (+40%)	+23.55
Equities (-40%)	-35.76

In addition to P&L, it is common practice to examine returns. Table 19 presents the portfolio returns under equity shock scenarios. A key observation is the asymmetric behavior of returns, reflecting the differing impact of positive and negative market movements.

Distribution-Based Stress Testing Results

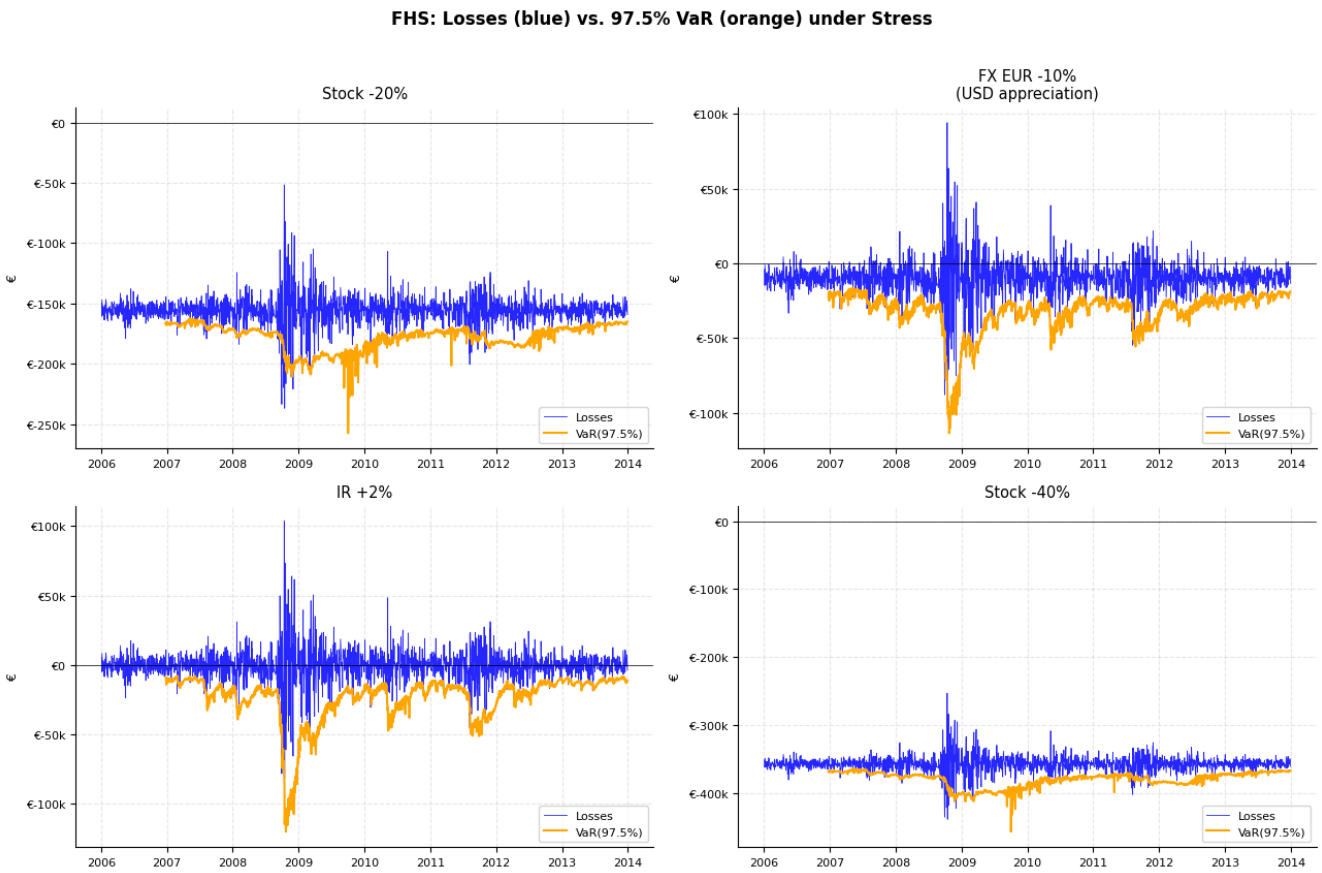

Figure 14: FHS simulation: portfolio losses (blue) and 1-day VaR (orange, 97.5%) under stress scenarios (equity -20%, FX EUR -10%, IR +2%, equity -40%).

Figure 14 displays the effects of all shocks on the different asset classes using the FHS framework. The blue line represents the P&L and the corresponding 97.5% VaR is shown in orange.

In the top left panel, we observe the equities shock of -20%. The shock causes the P&L to be deeply negative, fluctuating around a loss of approximately €150,000. In the top right panel, the FX EUR -10% shock has a considerably smaller effect on our P&L relative to equities. A notable volatility spike around the end of 2008 is observable, corresponding to the period when the EUR/USD exchange rate hit a record of 1.60 dollars per euro followed by a sharp correction. This caused our VaR to fall quite low during this period of high volatility.

The bottom left panel shows an interest rate increase of 2%. The VaR moves steadily alongside the volatility clusters of the P&L, indicating that interest rate exposure plays a minor role in our portfolio's short-term risk. The bottom right panel displays equities under a severe negative shock of -40%. Here the VaR flattens out, as losses of this magnitude already incorporate additional downward price action, diminishing the likelihood of further severe losses.

Interest Rate Sensitivity Analysis

Table 20 presents the analysis of interest rate changes on VaR and ES at the 99% confidence level using the FHS framework. As previously discussed, interest rates have a negligible effect on our portfolio risk metrics.

Table 20: IR Shocks — Impact on Portfolio VaR and ES (99%)

IR Shift	**VaR (%)**	**ES (%)**
+3bp	2.5621	2.9365
+2bp	2.5616	2.9360
-2bp	2.5596	2.9340
-3bp	2.5591	2.9335

To summarize from the stress test we conclude that equity risk is the main source of our portfolios' vulnerabilities. This is followed by exchange rate risk which poses a way smaller risk lastly and which forms a negligible amount of risk is the interest rate risk.